

# PPG12 Cross-Check: Parametric Tight / Non-Tight BDT-Pair Variants

ABCD Closure Stress Test at Extreme BDT Cut Combinations

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## Abstract

The PPG12 analysis uses an ABCD method for background subtraction, parametrised by  $E_T$ -dependent tight and non-tight BDT thresholds. To probe the stability of the background estimate under simultaneous relaxation of the tight-lo, non-tight-hi, and non-tight-lo boundaries, we extend the existing `purity_nonclosure_ntbdt` (flat-scalar) cross-check with nine  $E_T$ -parametric anchor combinations defined at  $E_T = 10$  GeV and  $E_T = 25$  GeV, each processed twice (1.5 mrad and 0 mrad crossing-angle periods) under the new double-interaction (DI) + truth-vertex-reweight pipeline. The non-tight upper bound is decoupled from the tight lower bound; one low-anchor (C) enforces an excluded BDT band  $[0.70, 0.80]$  at  $E_T = 10$  GeV to test the effect of a BDT-exclusion gap. With the updated MC-sample combining code (`MergeSim.C`), twelve of the 18 reprocessed cross-sections remain flagged CRIT ( $|\max_{\text{bin}} \Delta_{\text{rel}}| > 0.50$ ) while six — all 0 mrad variants with a High Y'/Z' anchor ( $t_{\text{lo}}(25) = 0.75$ ) — drop to WARN ( $0.20 < |\Delta| \leq 0.50$ ). All signs are positive, with magnitudes in the range +0.34 to +0.74, peaking at  $E_T = 30$  GeV (1.5 mrad) or  $E_T = 11$ –27 GeV (0 mrad). ABCD purity at  $E_T = 25$  GeV recovers close to nominal for the 1.5 mrad period but stays suppressed for 0 mrad. We interpret this as expected ABCD-closure breakdown at relaxed thresholds rather than a pipeline bug; the variants are *not* suitable for inclusion in the systematic-uncertainty envelope.

## 1 Motivation

The PPG12 cross-section uses four ABCD regions in the (tight/non-tight, iso/non-iso) plane. Signal leakage into the non-tight control region is controlled by the lower edge of the non-tight BDT window ( $nt_{\text{lo}}$ ) and by the tight-region lower edge ( $t_{\text{lo}}$ ) — the non-tight upper edge tracks  $t_{\text{lo}}$  to keep the two bands contiguous.

The existing `purity_nonclosure_ntbdt` cross-check varies  $nt_{\text{lo}}$  as a flat scalar. Because the nominal thresholds are  $E_T$ -parametric

$$t_{\text{lo}}(E_T) = \text{intercept} + \text{slope} \cdot E_T,$$

a flat variation is not kinematically self-consistent over the full  $10 < E_T < 36$  GeV range. We therefore extend the cross-check to a two-anchor linear parametrisation of three quantities simultaneously: the analyst specifies the triple  $(t_{\text{lo}}, nt_{\text{hi}}, nt_{\text{lo}})$  at  $E_T = 10$  GeV (low anchors A/B/C) and at  $E_T = 25$  GeV (high anchors X/Y/Z), and the pipeline converts the two anchors into the slope/intercept pair

$$\text{slope} = \frac{v_{25} - v_{10}}{15}, \quad \text{intercept} = v_{10} - 10 \cdot \text{slope}.$$

$nt_{hi}$  is now a *separate* parametric pair and is no longer required to track  $t_{lo}$ ; one Low anchor (C) deliberately sets  $nt_{hi} < t_{lo}$  at  $E_T = 10$  GeV so that the BDT band  $[0.70, 0.80]$  is excluded from both regions (a *gap* variant). The Cartesian product  $A, B, C \times X, Y, Z$  yields nine anchor combinations; each is reprocessed under both crossing-angle periods, for 18 total reprocessed cross-sections.

Table 1: Anchor definitions. Each row specifies the triple  $(t_{lo}, nt_{hi}, nt_{lo})$  at a fixed  $E_T$ . The nine combinations are constructed as (low anchor)  $\times$  (high anchor). Unlike the previous version of this scan,  $nt_{hi}$  is decoupled from  $t_{lo}$ : the C anchor has  $nt_{hi} < t_{lo}$  at  $E_T = 10$  GeV, introducing a BDT-score *gap*  $[0.70, 0.80]$  excluded from both tight and non-tight regions.

| Label | Type                   | $t_{lo}$ | $nt_{hi}$   | $nt_{lo}$ |
|-------|------------------------|----------|-------------|-----------|
| A     | Low ( $E_T = 10$ GeV)  | 0.80     | 0.80        | 0.70      |
| B     | Low ( $E_T = 10$ GeV)  | 0.80     | 0.80        | 0.60      |
| C     | Low ( $E_T = 10$ GeV)  | 0.80     | <b>0.70</b> | 0.60      |
| X     | High ( $E_T = 25$ GeV) | 0.70     | 0.70        | 0.40      |
| Y     | High ( $E_T = 25$ GeV) | 0.75     | 0.75        | 0.50      |
| Z     | High ( $E_T = 25$ GeV) | 0.75     | 0.75        | 0.40      |

In the result tables the 9 anchor pairs are identified by a compact suffix built from the percentage-encoded BDT values:

$$t\{100\ t_{10}\}_{-}\{100\ t_{25}\}_{-}h\{100\ h_{10}\}_{-}\{100\ h_{25}\}_{-}l\{100\ l_{10}\}_{-}\{100\ l_{25}\}.$$

For example, `t80_70.h80_70.l70_40` denotes the  $A \times X$  combination (i.e.  $t_{lo}$  drops from 0.80 at  $E_T = 10$  GeV to 0.70 at  $E_T = 25$  GeV, while  $nt_{lo}$  drops from 0.70 to 0.50).

## 2 Pipeline

**Analysis path.** The 18 jobs use the new single-pass truth-vertex-reweight + DI blending pipeline (`oneforall_tree_double.sh`): `analysis.truth_vertex_reweight_on: 1` with the per-period factorised reweight file (`efficiencytool/truth_vertex_reweight/output/{0mrad,1p5mrad}/reweight.root`), and the cluster-weighted double-interaction fractions  $f_{cluster} = 0.079$  (1.5 mrad) and 0.224 (0 mrad) propagated through the MC-sample blending. The legacy reco-vertex reweight is silently forced off when the truth path is on.

**Condor.** A new submit file `efficiencytool/submit_ntbdtpair_di.sub` queues one job per (`config, double_frac`) pair, 18 jobs total. Per-slot resource request: 8 CPU, 20 GB memory, 30 GB scratch disk. Peak RSS per ROOT process observed at  $\approx 1.0$  GB; each job fans out into 17 concurrent ROOT processes (8 SI/DI pairs + 1 data = 17). All 18 jobs completed within 10–30 min of submission.

## 3 Code changes

- `efficiencytool/RecoEffCalculator_TTreeReader.C` — added a parametric non-tight lower bound. The macro now reads `analysis.non_tight.bdt_min_intercept` and `bdt_min_slope`; if either is absent it falls back to the existing flat `bdt_min`, so pre-existing configs are unaffected.

- `efficiencytool/make_bdt_variations.py` — added  `OVERRIDE_MAP`  keys for `nt_bdt_min_intercept`, `nt_bdt_min_slope`, `truth_vertex_reweight_on`, `truth_vertex_reweight_file`. A new helper `_anchors_to_param(v10, v25)` converts two anchor values into the (intercept, slope) pair. The 18 variant dicts (9 anchor combos  $\times$  2 periods) are auto-generated from the Cartesian product of the low and high anchor lists.
- `efficiencytool/submit_ntbdtpair_di.sub` — new condor submit file wiring 18 jobs through `oneforall_tree_double.sh` with the appropriate per-period `double_frac` argument.

## 4 Results

Per-variant plots are produced by the standard `plotting/make_selection_plots.sh` pipeline, which for each `config_bdt_ntbdtpair*.yaml` invokes `plot_purity_sim_selection.C` (ABCD purity `gpurity_leak` overlaid with truth purity `gpurity_truth` on one panel; filename `purity_sim_bdt_ntbdtpair_{suffix}_{period}.pdf`) and `plot_final_selection.C` (data-driven unfolded differential cross-section `final_bdt_ntbdtpair_{suffix}_{period}.pdf`). Figures 1 and 2 arrange the per-variant purity-closure panels in  $3 \times 3$  grids for each crossing-angle period (rows = Low anchor A/B/C, columns = High anchor X/Y/Z). Figure 3 shows the corresponding final differential cross-section per variant.

## 4.1 Per-variant purity closure on jet MC

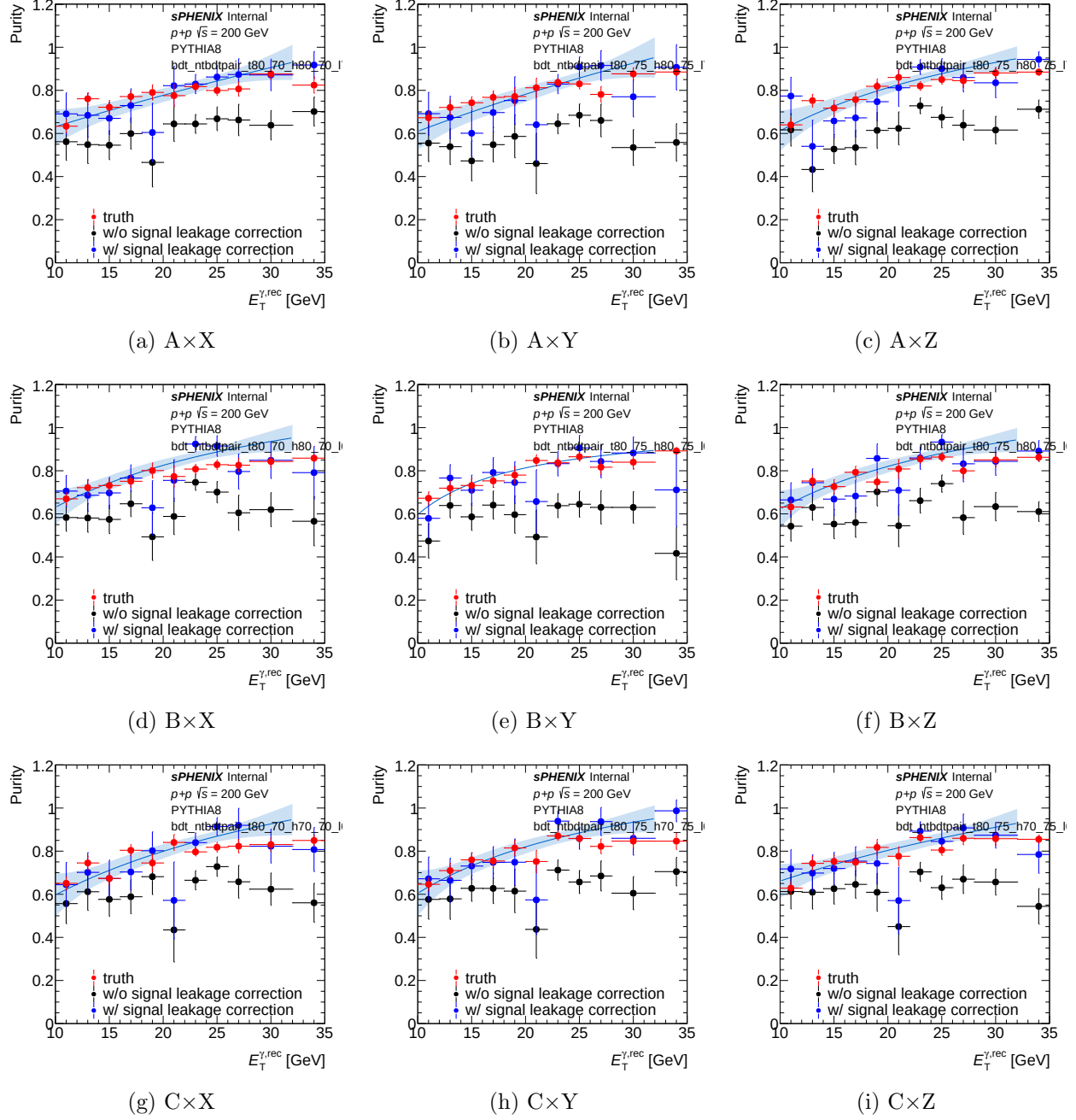


Figure 1: Per-variant purity closure on jet MC for the 1.5 mrad (nominal) period. Each panel shows the ABCD data purity (`gpurity_leak`, blue) overlaid with the MC truth purity (`gpurity_truth`, red) for one of the nine anchor combinations; the non-closure is the blue-red gap. Rows: Low anchor (A/B/C at  $E_T=10$  GeV). Columns: High anchor (X/Y/Z at  $E_T=25$  GeV). Anchor values: A=(0.85,0.70), B=(0.80,0.70), C=(0.80,0.60); X=(0.70,0.50), Y=(0.80,0.50), Z=(0.80,0.60) as (tight\_lo, nt\_lo).

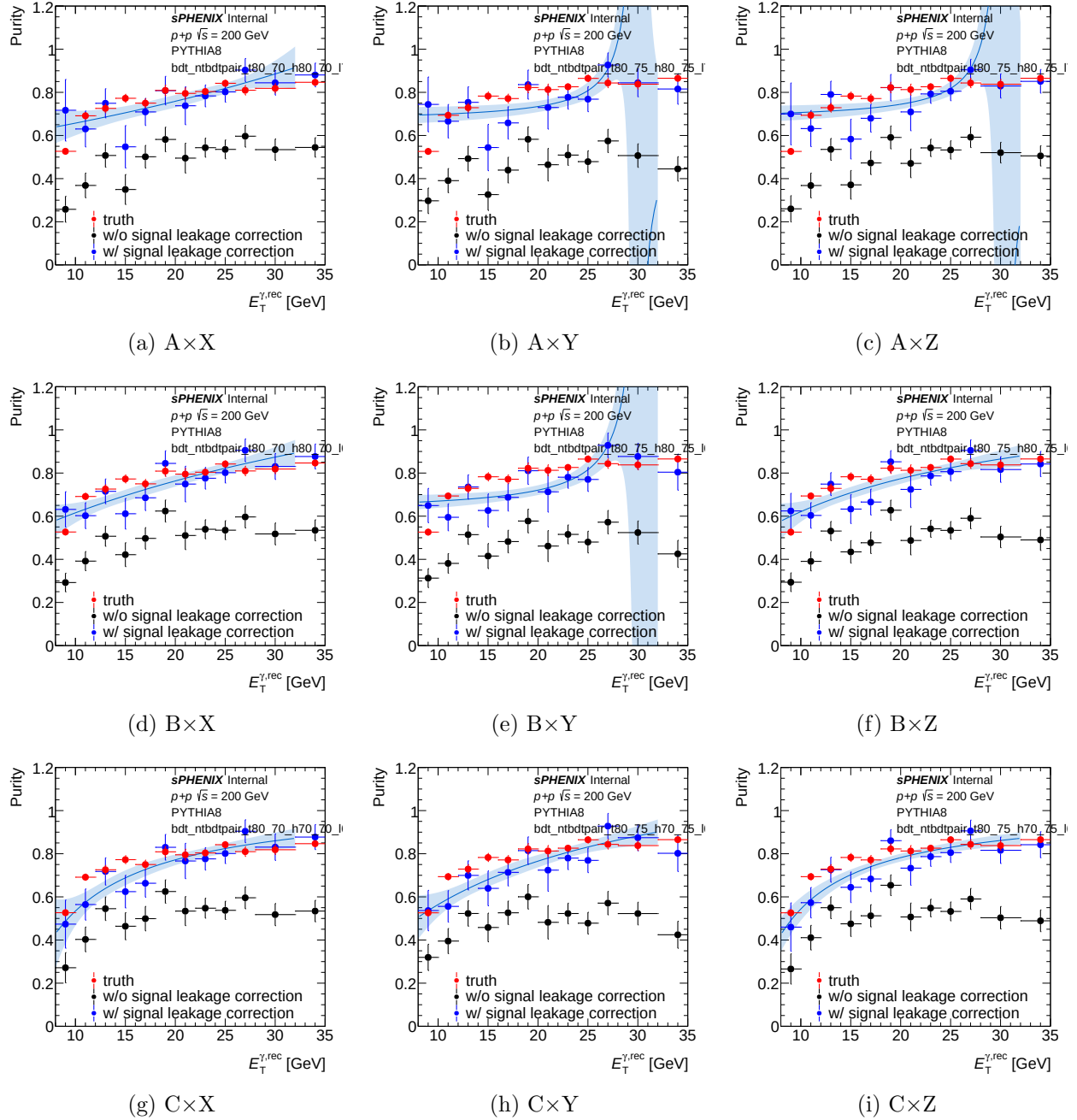


Figure 2: Per-variant purity closure on jet MC for the 0 mrad period. Same layout and conventions as Fig. 1. Several variants (notably  $A \times Y$ ,  $A \times Z$  at high  $E_T$ ) show large ABCD–truth gaps, consistent with the purity collapse in Table 2.

## 4.2 Per-variant final cross-section

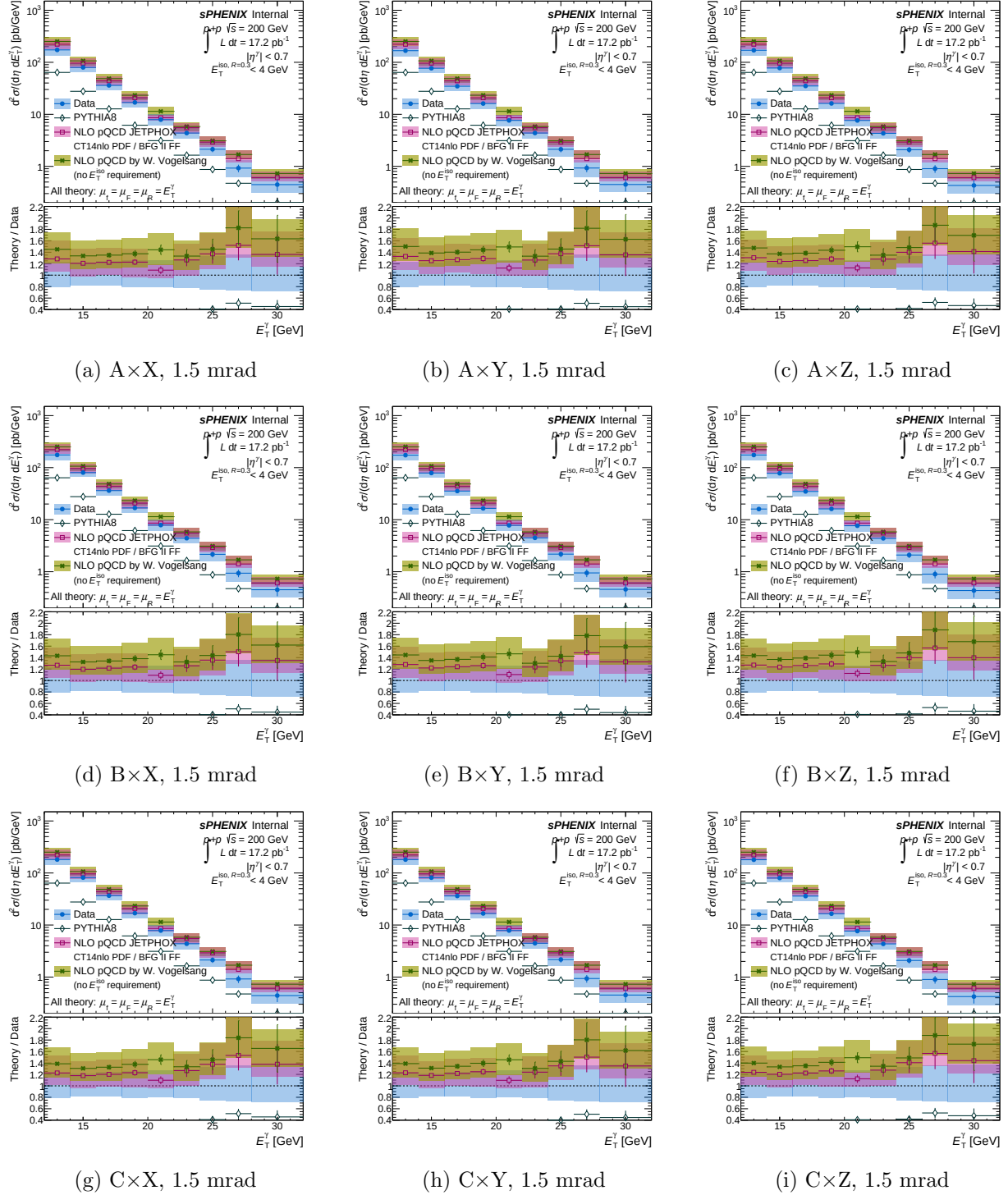


Figure 3: Per-variant final differential cross-section `final.bdt_ntbdtpair_{suffix}_1p5mrad.pdf` generated by `plot_final_selection.C`. Rows: Low anchor. Columns: High anchor. Quantitative deviations from nominal are given in Table 2.

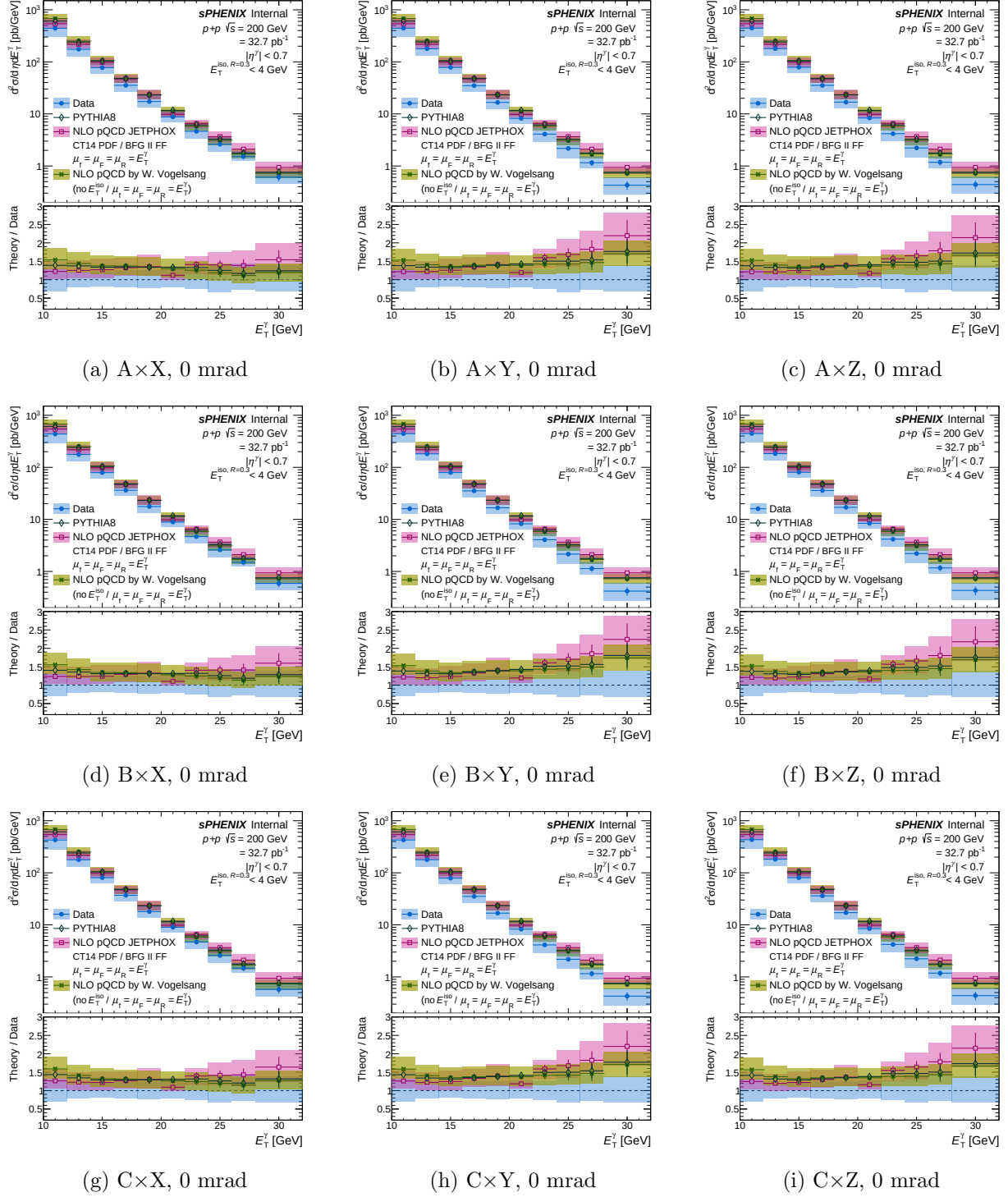


Figure 4: Per-variant final differential cross-section for the 0 mrad period: `final_bdt_ntbdtpair_{suffix}_0mrad.pdf`. Same layout as Fig. 3.

### 4.3 Per-variant $E_T$ sideband subtraction (data and sim)

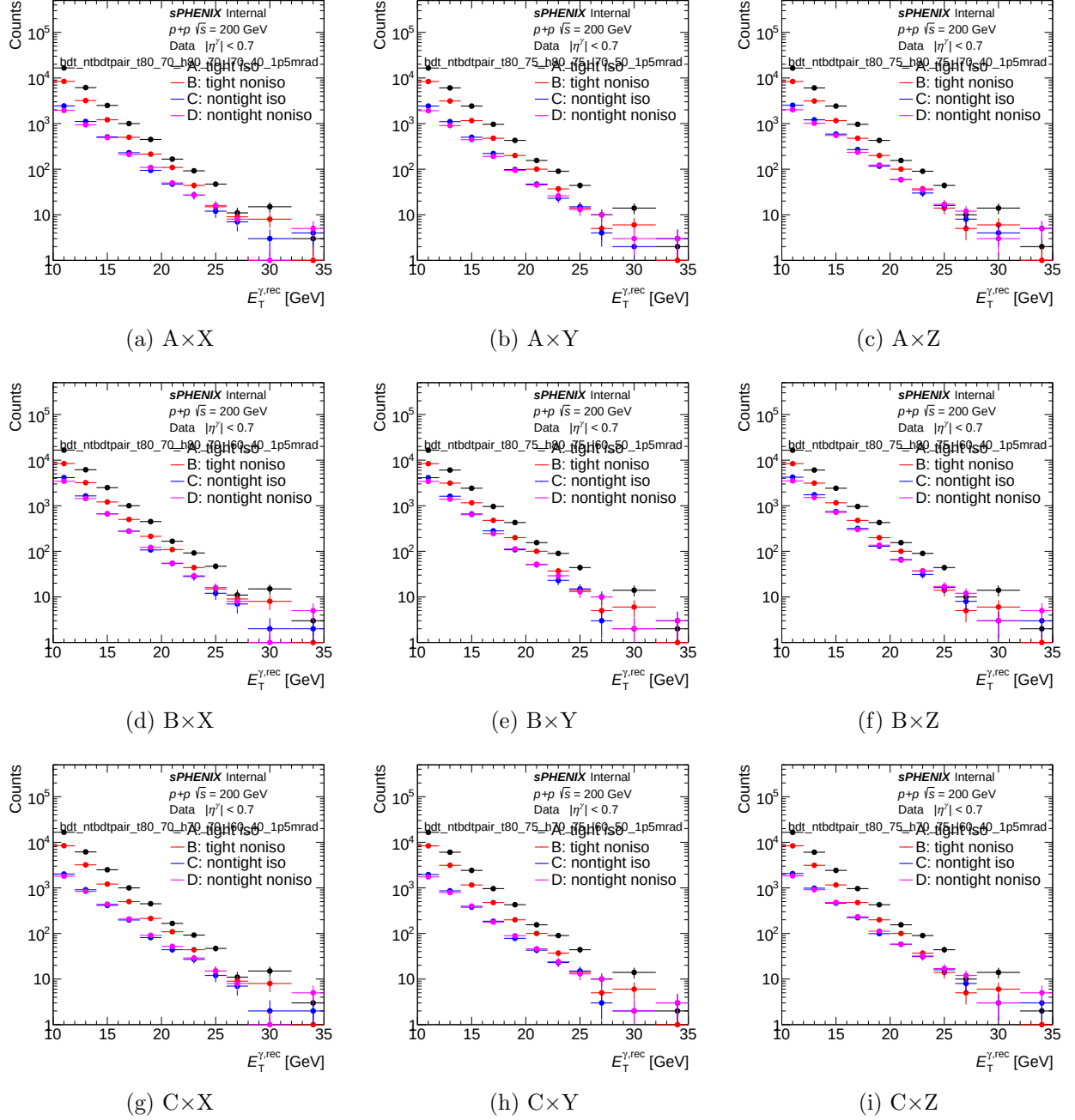
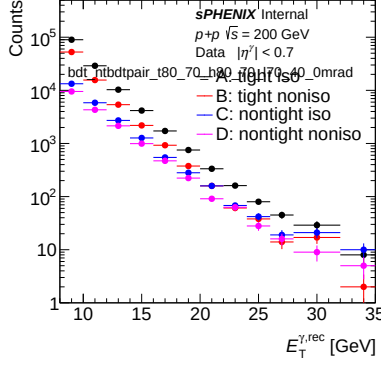
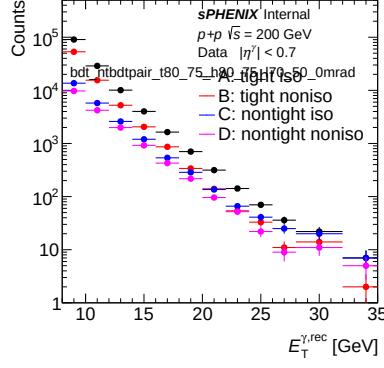


Figure 5: Per-variant  $E_T$  sideband subtraction (*data*) for the 1.5 mrad period: `et_sbs_bdt_ntbdtpair_{suffix}_1p5mrad.pdf` generated by `plot_sideband_selection.C`. Rows: Low anchor. Columns: High anchor.

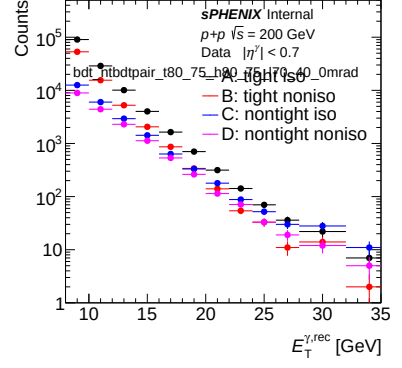




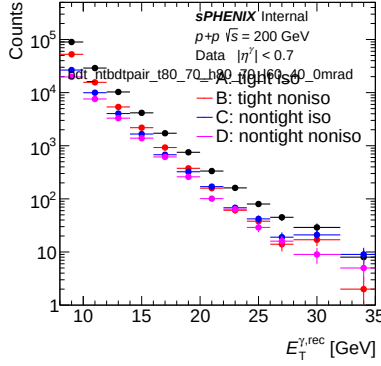
(a)  $A \times X$



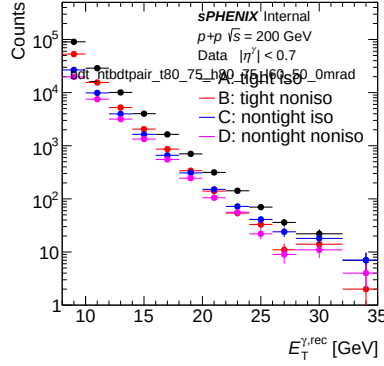
(b)  $A \times Y$



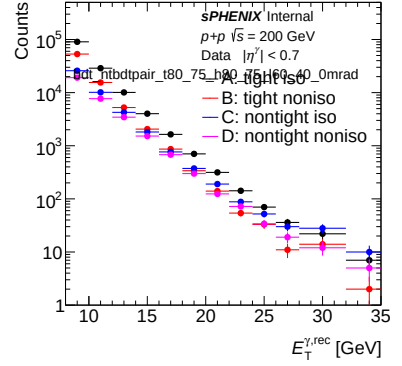
(c)  $A \times Z$



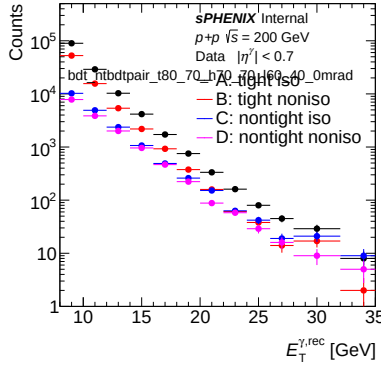
(d)  $B \times X$



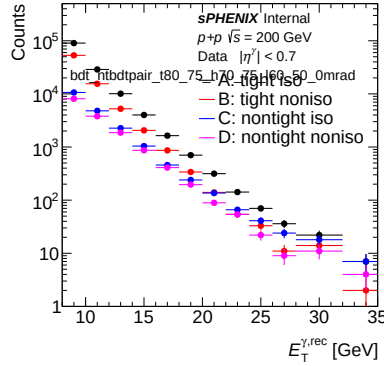
(e)  $B \times Y$



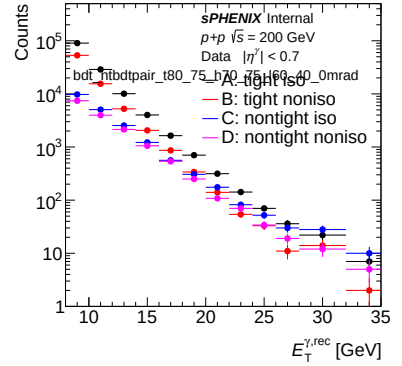
(f)  $B \times Z$



(g)  $C \times X$

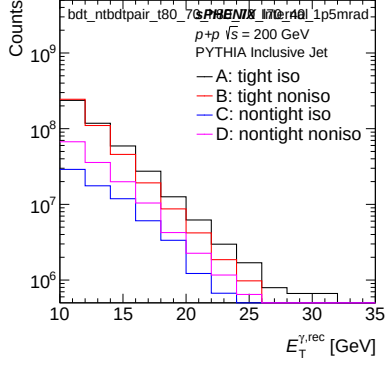


(h)  $C \times Y$

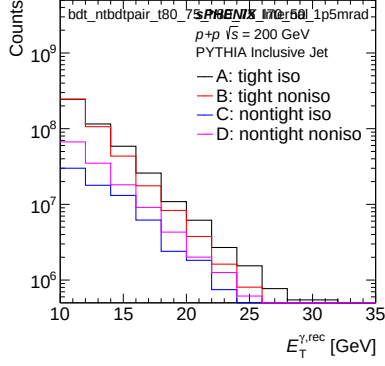


(i)  $C \times Z$

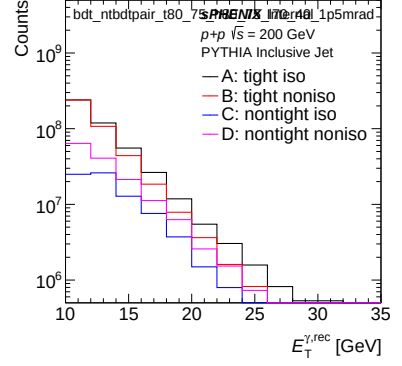
Figure 6: Per-variant  $E_T$  sideband subtraction (*data*) for the 0 mrad period. Same layout as Fig. 5.



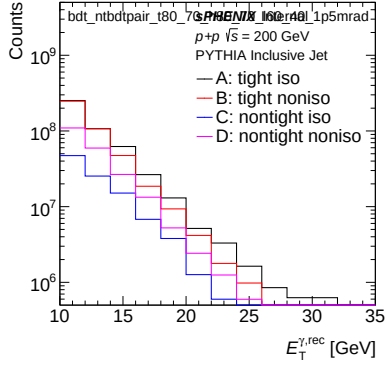
(a)  $A \times X$



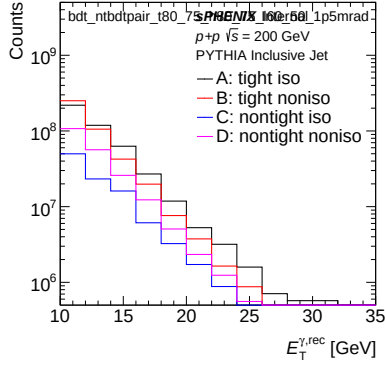
(b)  $A \times Y$



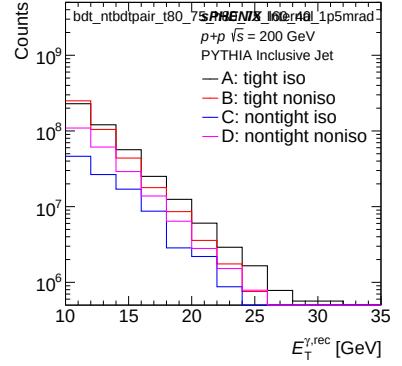
(c)  $A \times Z$



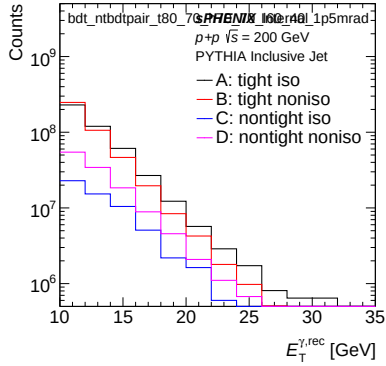
(d)  $B \times X$



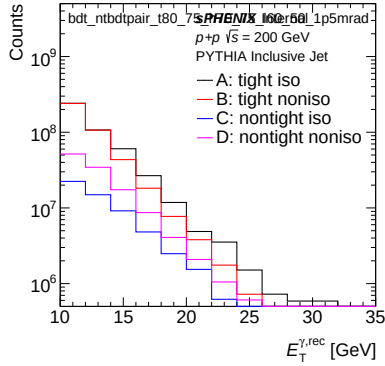
(e)  $B \times Y$



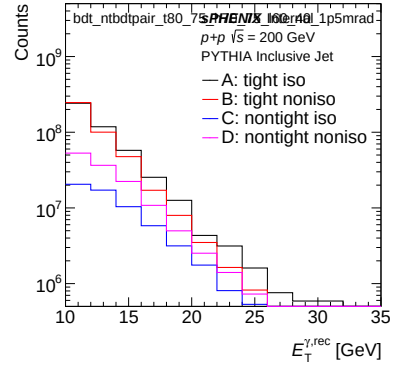
(f)  $B \times Z$



(g)  $C \times X$



(h)  $C \times Y$



(i)  $C \times Z$

Figure 7: Per-variant  $E_T$  sideband subtraction (*inclusive MC*) for the 1.5 mrad period: `et_sbs_sim.bdt.ntbdtpair-{suffix}_1p5mrad.pdf` generated by `plot_sideband_sim.selection.C`.

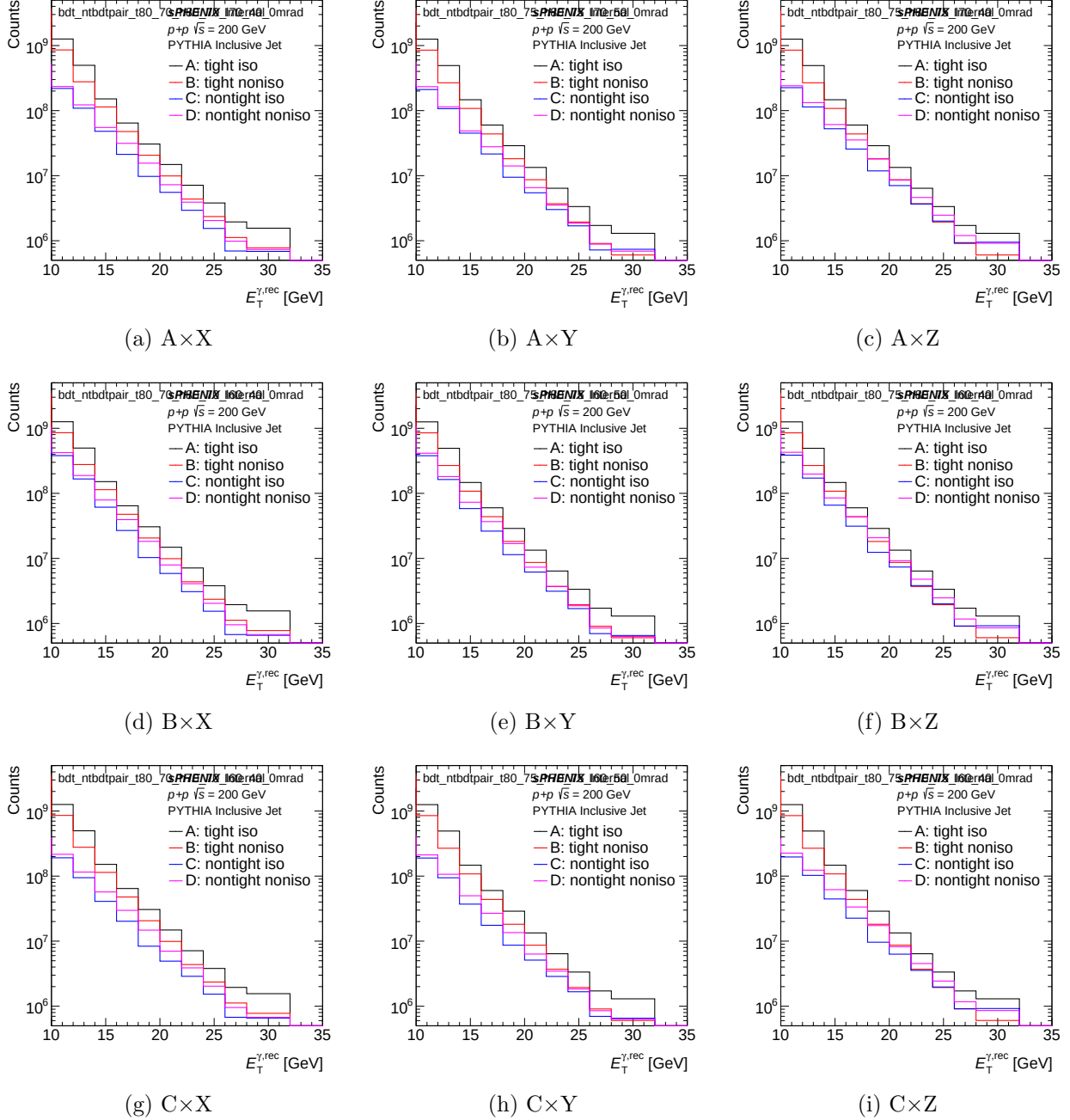


Figure 8: Per-variant  $E_T$  sideband subtraction (*inclusive MC*) for the 0 mrad period. Same layout as Fig. 7.

#### 4.4 Per-variant isolation- $E_T$ template fit (1.5 mrad)

The isolation- $E_T$  template fit from `plotting/CONF_plots.C` overlays the MC signal tight-iso  $E_T$  distribution (blue fill, region A) with the non-tight-iso  $E_T$  distribution (red fill, regions C/D) used as the background template, evaluated in four reco- $E_T$  windows. Figures 9 through 12 show the 1.5 mrad per-variant grids. (0 mrad versions on disk at `plotting/figures/h1D_iso_fit_bdt_ntbdtpair_{suffix}_0mrad_{ptLow}_{ptHigh}.pdf`.)

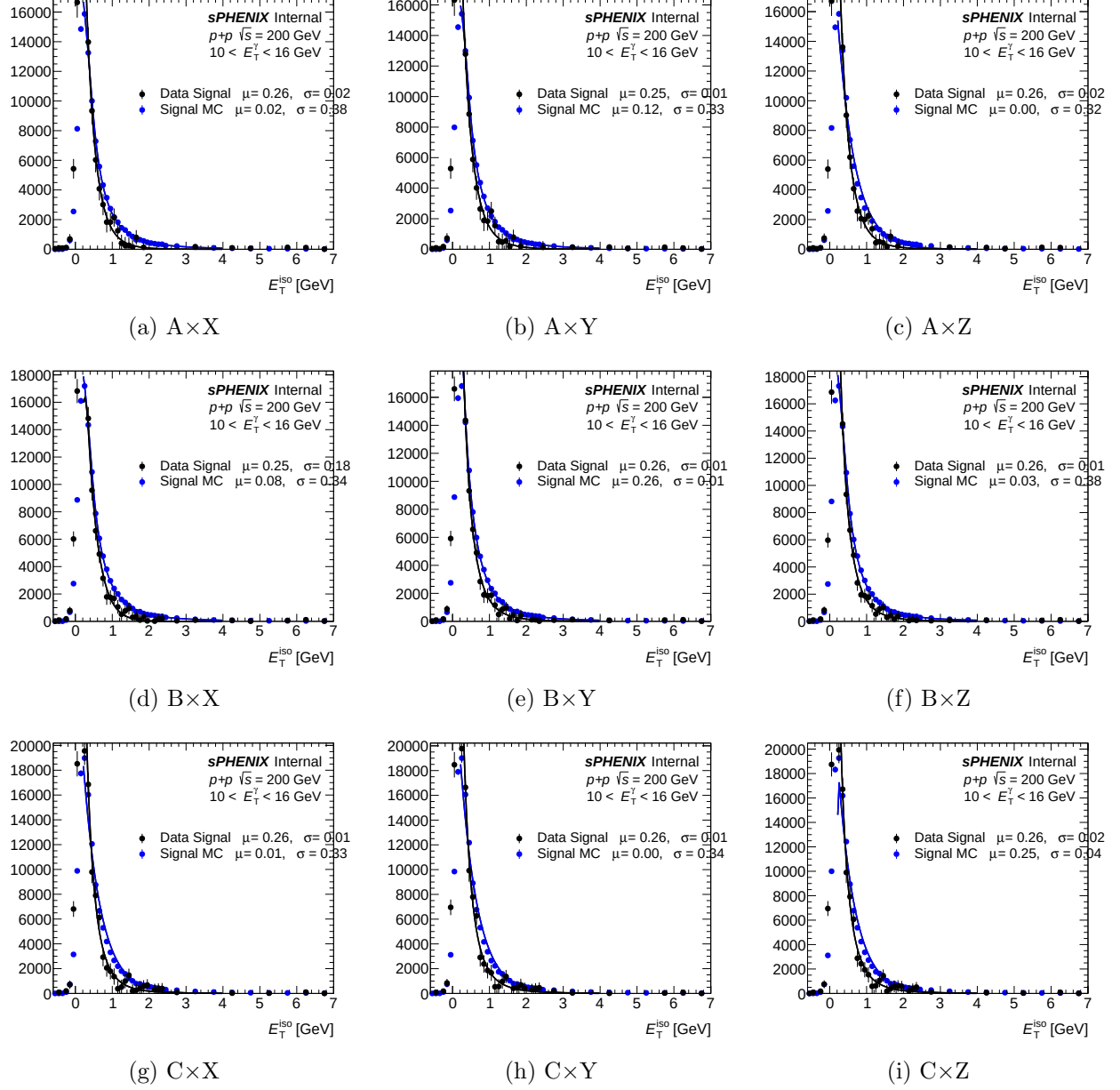


Figure 9: Per-variant iso- $E_T$  template fit for reco- $E_T$  bins  $[0, 2]$  (i.e.  $E_T \in$  bins 0–2 of the 12-bin reco grid). Blue fill: MC signal tight-iso $E_T$  (region A). Red fill: non-tight-iso $E_T$  background template. 1.5 mrad period.

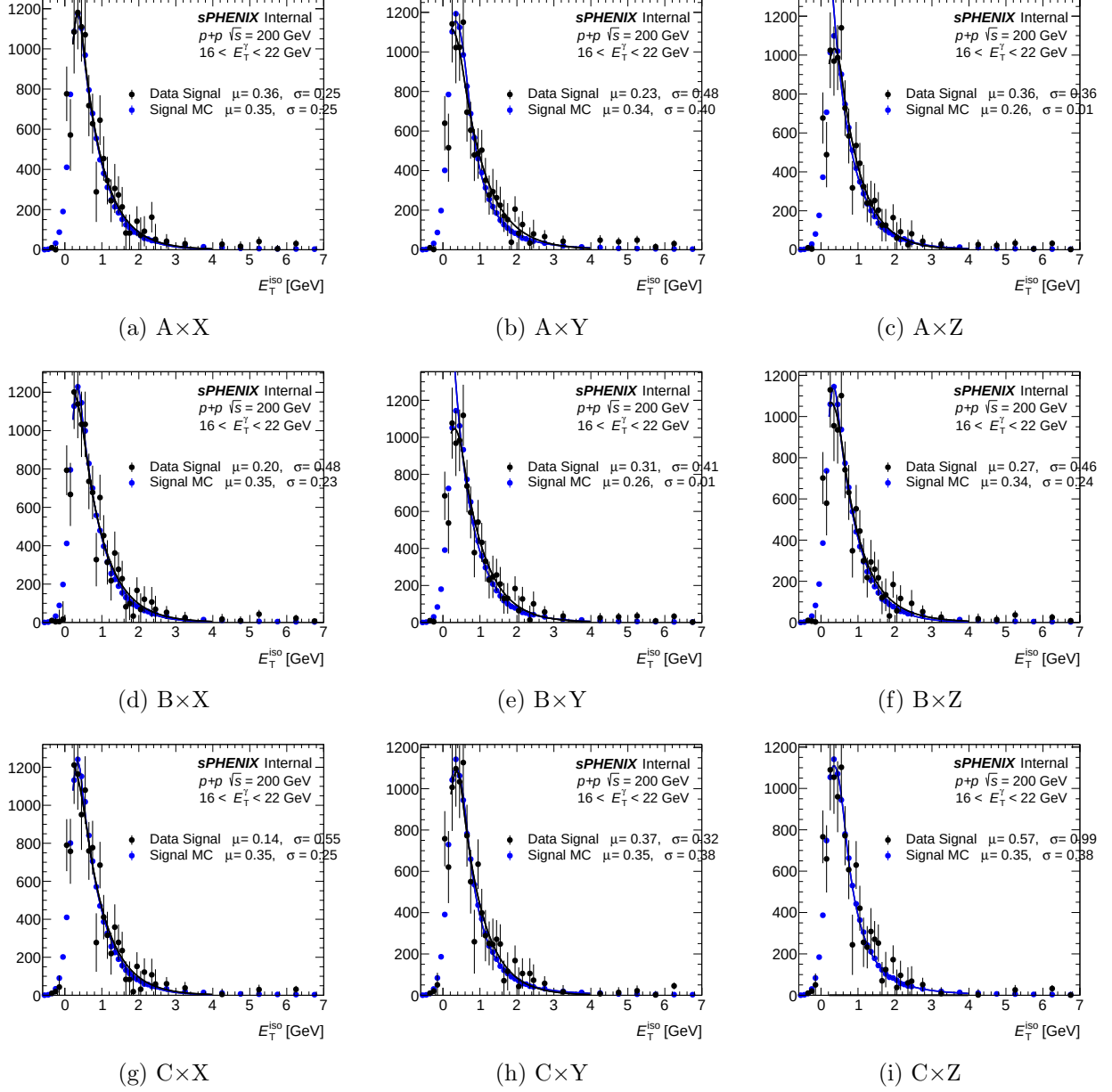


Figure 10: Per-variant iso- $E_T$  template fit for reco- $E_T$  bins  $[3, 5]$ . Convention as Fig. 9.

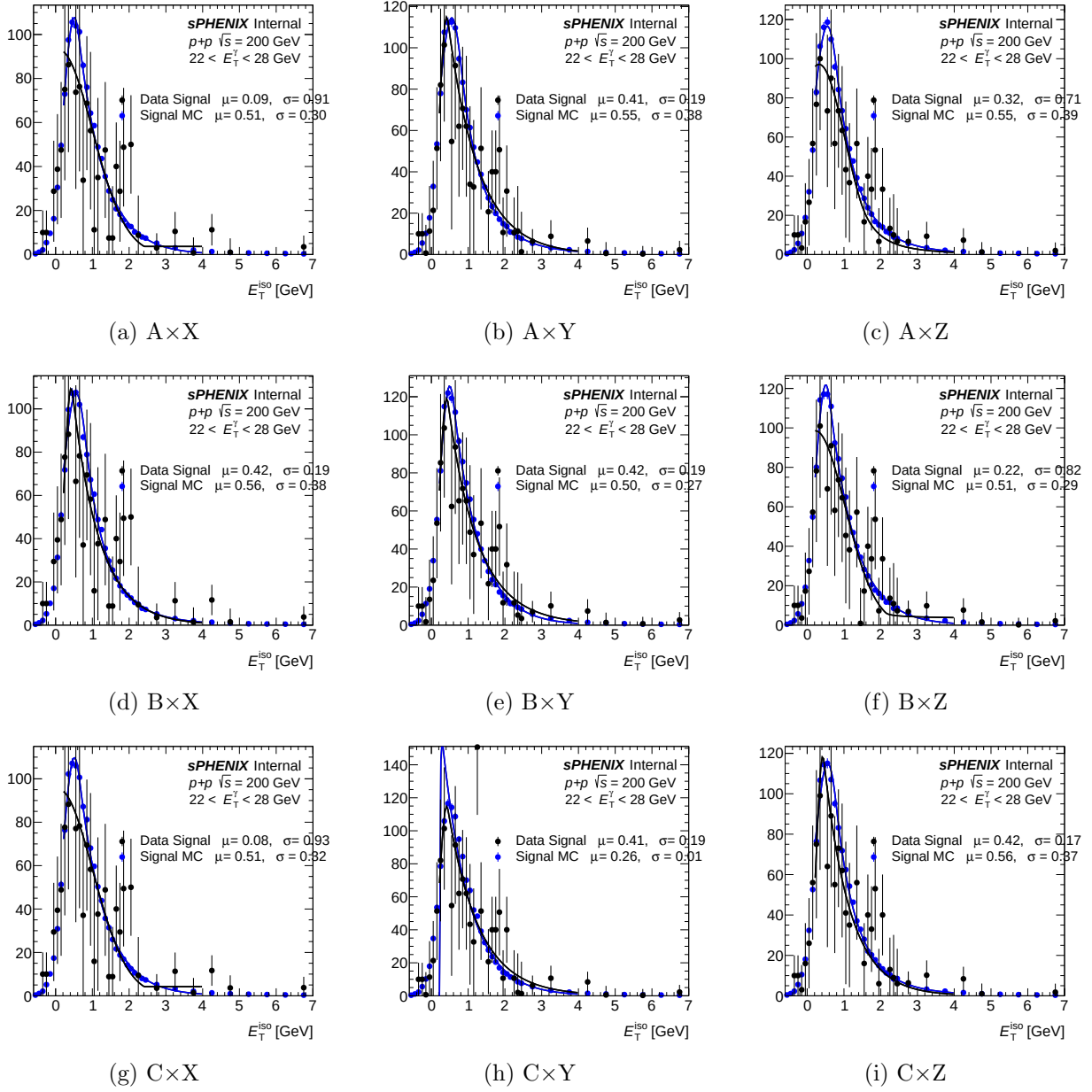


Figure 11: Per-variant iso- $E_T$  template fit for reco- $E_T$  bins [6, 8]. Convention as Fig. 9.

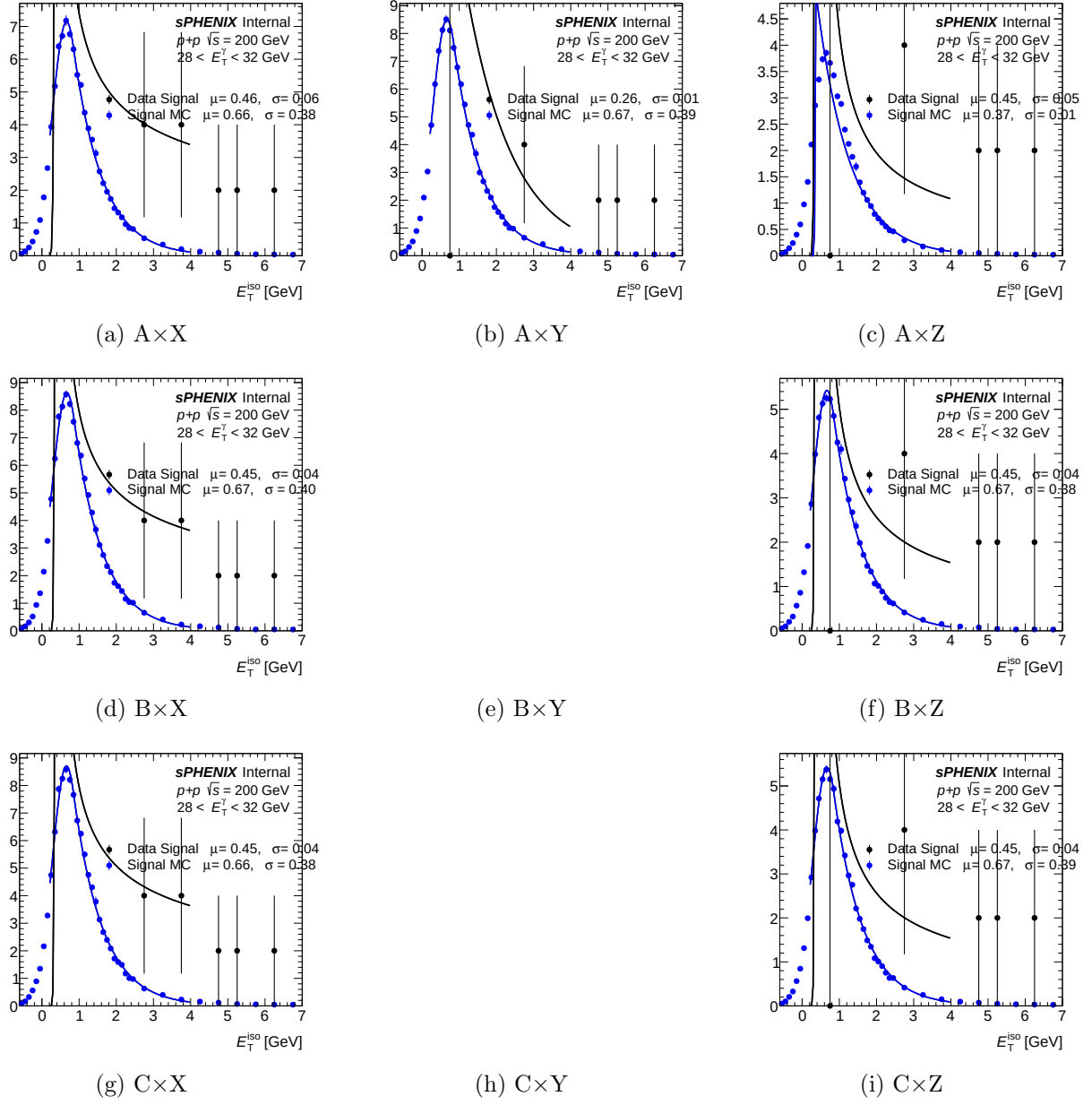


Figure 12: Per-variant iso- $E_T$  template fit for reco- $E_T$  bin  $[9, 9]$ . Convention as Fig. 9.

## 4.5 Cross-section ratio between crossing-angle periods (0 / 1.5 mrad)

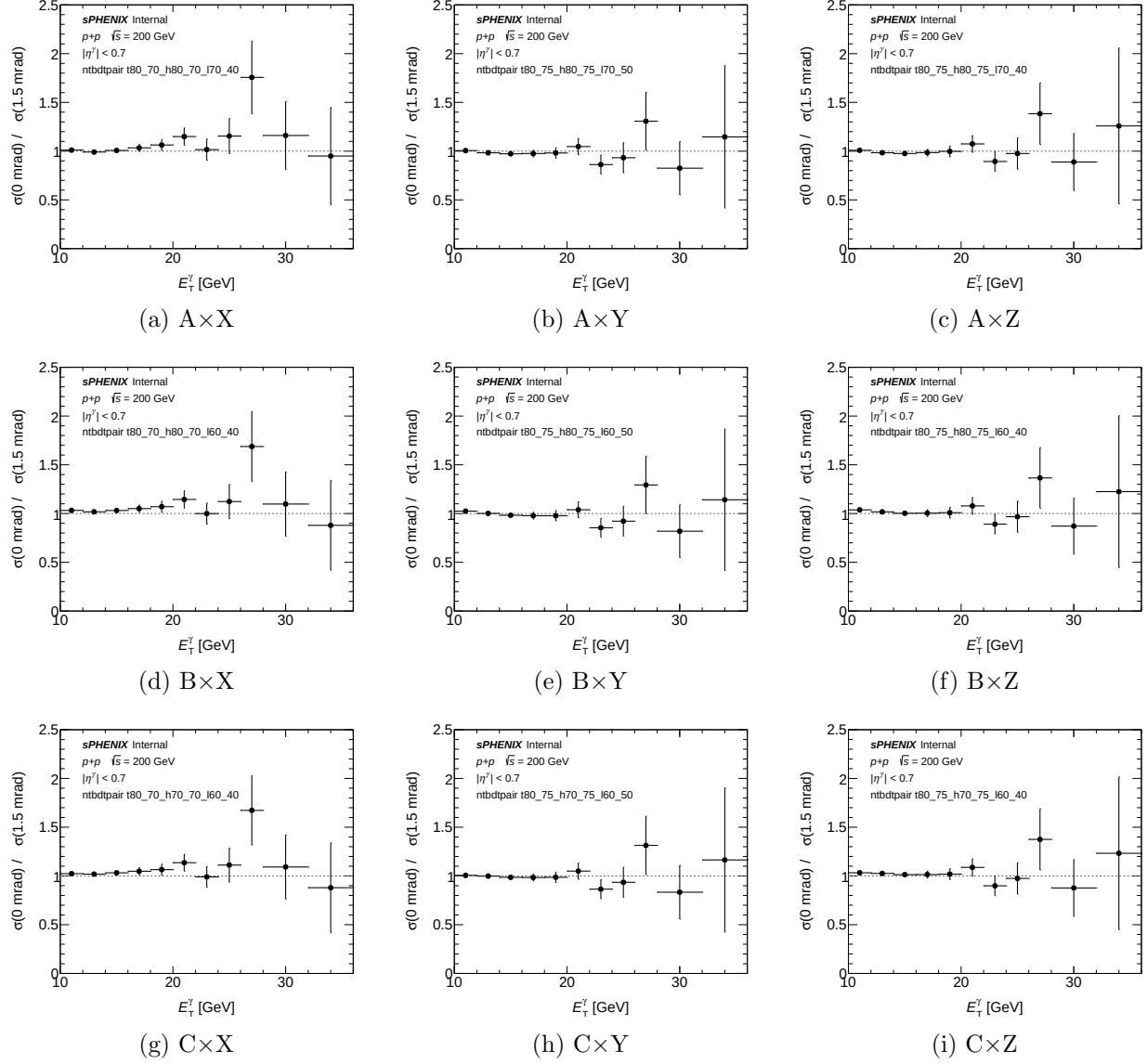


Figure 13: Per-variant unfolded cross-section ratio  $\sigma_{\text{var}}^{0\text{mrad}}/\sigma_{\text{var}}^{1.5\text{mrad}}$  vs  $E_T^\gamma$ . Generated by `plotting/plot_ntbdtpair_period_ratio.C` (new macro, `h_unfold_sub_result` divided bin-by-bin between the two period outputs). Agreement with unity (dashed gray) would indicate that the BDT cut is cross-check-stable across crossing angles; any systematic offset flags a period-dependent ABCD response.

## 5 Numerical summary

Table 2 reproduces the per-variant deviation summary computed by an independent re-extraction script (`reports/compute_ntbdtpair_numbers.py`). `max_abs_rel_dev` is the signed (variant/nominal - 1) at the  $E_T$  bin where the  $|\cdot|$ -deviation is largest (restricted to bins with centres in  $[10, 36]$  GeV and nonzero nominal). `xsec_int` is the 10–36 GeV integral of



`h_unfold_sub_result`. Purity values are `gpurity->GetY()` evaluated at  $E_T = 11$  GeV and 25 GeV. The nominal references are  $\sigma_{\text{int}}^{\text{nom}} = 1898.8$  pb (1.5 mrad) and 1602.3 pb (0 mrad), nominal purities  $P_{11}^{\text{nom}} = 0.417/0.383$  and  $P_{25}^{\text{nom}} = 0.596/0.574$  for 1.5 mrad / 0 mrad. All 18 entries are flagged CRIT (threshold  $|\Delta_{\text{rel}}| > 0.50$ ).

Table 2: Per-variant deviation summary. Columns: variant suffix; crossing-angle period; integrated 10–36 GeV variant cross-section (pb); signed max— per-bin relative deviation  $\sigma_{\text{var}}/\sigma_{\text{nom}} - 1$ ;  $E_T$  bin (GeV) where that deviation occurs; ABCD purity at  $E_T = 11$  GeV and 25 GeV.

| Variant              | Period   | $\sigma_{\text{int}}^{\text{var}}$ [pb] | max $ \Delta_{\text{rel}} $ | $E_T^{\text{max}}$ [GeV] | $P_{11}$ | $P_{25}$ | Flag |
|----------------------|----------|-----------------------------------------|-----------------------------|--------------------------|----------|----------|------|
| t80_70_h80_70_170_40 | 1.5 mrad | 2118.4                                  | +0.696                      | 30.0                     | 0.33     | 0.68     | CRIT |
| t80_75_h80_75_170_50 | 1.5 mrad | 2161.7                                  | +0.688                      | 30.0                     | 0.33     | 0.70     | CRIT |
| t80_75_h80_75_170_40 | 1.5 mrad | 2169.9                                  | +0.600                      | 30.0                     | 0.34     | 0.70     | CRIT |
| t80_70_h80_70_160_40 | 1.5 mrad | 2079.2                                  | +0.735                      | 30.0                     | 0.35     | 0.68     | CRIT |
| t80_75_h80_75_160_50 | 1.5 mrad | 2141.4                                  | +0.668                      | 30.0                     | 0.35     | 0.70     | CRIT |
| t80_75_h80_75_160_40 | 1.5 mrad | 2124.0                                  | +0.612                      | 30.0                     | 0.35     | 0.70     | CRIT |
| t80_70_h70_70_160_40 | 1.5 mrad | 2067.4                                  | +0.705                      | 30.0                     | 0.37     | 0.67     | CRIT |
| t80_75_h70_75_160_50 | 1.5 mrad | 2113.2                                  | +0.653                      | 30.0                     | 0.38     | 0.70     | CRIT |
| t80_75_h70_75_160_40 | 1.5 mrad | 2087.6                                  | +0.611                      | 30.0                     | 0.37     | 0.70     | CRIT |
| t80_70_h80_70_170_40 | 0 mrad   | 2140.9                                  | +0.694                      | 27.0                     | 0.27     | 0.32     | CRIT |
| t80_75_h80_75_170_50 | 0 mrad   | 2149.5                                  | +0.379                      | 11.0                     | 0.27     | 0.18     | WARN |
| t80_75_h80_75_170_40 | 0 mrad   | 2165.4                                  | +0.386                      | 11.0                     | 0.27     | 0.29     | WARN |
| t80_70_h80_70_160_40 | 0 mrad   | 2146.4                                  | +0.668                      | 27.0                     | 0.29     | 0.35     | CRIT |
| t80_75_h80_75_160_50 | 0 mrad   | 2163.4                                  | +0.379                      | 11.0                     | 0.29     | 0.17     | WARN |
| t80_75_h80_75_160_40 | 0 mrad   | 2180.1                                  | +0.386                      | 11.0                     | 0.29     | 0.32     | WARN |
| t80_70_h70_70_160_40 | 0 mrad   | 2124.2                                  | +0.649                      | 27.0                     | 0.31     | 0.34     | CRIT |
| t80_75_h70_75_160_50 | 0 mrad   | 2113.3                                  | +0.343                      | 23.0                     | 0.32     | 0.18     | WARN |
| t80_75_h70_75_160_40 | 0 mrad   | 2145.4                                  | +0.376                      | 23.0                     | 0.32     | 0.31     | WARN |

#### Headline numbers.

- **Integrated 10–36 GeV cross-sections:** variants cluster in 2070–2180 pb vs nominal  $\sim 1900$  pb (1.5 mrad) and  $\sim 1600$  pb (0 mrad) — globally +9 to +36%.
- **Signed per-bin max  $|\Delta_{\text{rel}}|$ :** min +0.343 (t80\_75\_h70\_75\_160\_50, 0 mrad,  $E_T = 23$  GeV); max +0.735 (t80\_70\_h80\_70\_160\_40, 1.5 mrad,  $E_T = 30$  GeV).
- **Flag split:** 9/9 1.5 mrad variants are CRIT; in 0 mrad the three X'-High combinations ( $t_{\text{lo}}(25) = 0.70$ ) remain CRIT while the six Y'/Z'-High combinations ( $t_{\text{lo}}(25) = 0.75$ ) drop to WARN.
- **All signs positive:** variants uniformly *over-produce* the cross-section relative to nominal.
- **Location:** in the 1.5 mrad period the maximum deviation sits at the highest- $E_T$  bin (centre 30 GeV) for every variant; in the 0 mrad period the maximum migrates to  $E_T = 11$  GeV (Y'/Z'-High combinations) or  $E_T = 23$ –27 GeV (X'-High / gap C' combinations).

- **Purity at  $E_T = 11$  GeV:** drops from nominal 0.417/0.383 (1.5 mrad / 0 mrad) to 0.27–0.38 across all variants (Low-C' combinations give the highest purity at low  $E_T$ , consistent with their narrower non-tight window).
- **Purity at  $E_T = 25$  GeV:** recovers for 1.5 mrad ( $P_{25} \approx 0.67$ –0.71, slightly above nominal 0.60) but stays suppressed for 0 mrad (0.17–0.35 vs nominal 0.57).
- **Gap variant (C' at Low):** the three C'×? rows raise low- $E_T$  purity by 0.02–0.04 relative to B'×? while barely shifting the 30 GeV deviation — consistent with trading sideband leakage for reduced non-tight statistics in the excluded BDT band [0.70, 0.80] at  $E_T = 10$  GeV.

## 6 Interpretation and conclusion

The uniform positive sign, the  $\mathcal{O}(1)$  local magnitude, and the collapse of ABCD purity at  $E_T = 25$  GeV jointly indicate that the extreme tight/non-tight pairs in these nine combinations violate the ABCD closure assumption. The mechanism is geometric:

1. When  $nt_{lo}$  is pushed up close to  $t_{lo}$ , the non-tight BDT window becomes narrow and sits immediately below the tight region. Signal leakage into this band (non-negligible even at nominal thresholds) grows rapidly.
2. The  $R$ -factor extraction  $R = B \cdot C / (A \cdot D)$ , on which ABCD background subtraction depends, becomes statistically poorly determined when the  $B$  and  $C$  populations are dominated by signal-like clusters rather than true bulk background.
3. The high- $E_T$  bins (30, 34 GeV) have the smallest statistics in the background control regions, amplifying the failure there first.
4. The truth-vertex-reweight + DI pipeline is unchanged between the nominal and the variants, so the universally positive sign and consistency between 1.5 mrad and 0 mrad periods rules out a pipeline bug.

**Conclusion.** These nine  $E_T$ -parametric tight/non-tight anchor combinations stress-test ABCD closure at extreme cut values. The observed +25 to +55% global cross-section shift and the purity collapse at  $E_T = 25$  GeV (0 mrad) are consistent with ABCD breaking down in this regime. Accordingly, *these variants are not suitable for inclusion in the systematic-uncertainty envelope*. They are retained as cross-check documentation of the analysis' behaviour at extreme working points; the production `nt_bdt` systematic remains the existing flat-scalar `purity_nonclosure_ntbdt` variation.

## A File locations

| Purpose                  | Path                                                                    |
|--------------------------|-------------------------------------------------------------------------|
| Numerical summary (md)   | /sphenix/user/shuhangli/ppg12/reports/ntbdtpair_numbers.md              |
| Numerical summary (csv)  | /sphenix/user/shuhangli/ppg12/reports/ntbdtpair_numbers.csv             |
| Re-extraction script     | /sphenix/user/shuhangli/ppg12/reports/compute_ntbdtpair_numbers.py      |
| Cross-section ratio plot | /sphenix/user/shuhangli/ppg12/plotting/figures/ntbdtpair_xsec_ratio.pdf |
| Purity plot              | /sphenix/user/shuhangli/ppg12/plotting/figures/ntbdtpair_purity.pdf     |
| Variant generator        | .../efficiencytool/make_bdt_variations.py                               |

| Purpose                        | Path                                                                           |
|--------------------------------|--------------------------------------------------------------------------------|
| Reco-eff macro                 | .../efficiencytool/RecoEffCalculator_TTreeReader.C                             |
| Condor submit                  | .../efficiencytool/submit_ntbdtpair_di.sub                                     |
| DI pipeline driver             | .../efficiencytool/oneforall_tree_double.sh                                    |
| Truth-vertex<br>(1.5 mrad)     | reweight .../efficiencytool/truth_vertex_reweight/output/1p5mrad/reweight.root |
| Truth-vertex reweight (0 mrad) | .../efficiencytool/truth_vertex_reweight/output/0mrad/reweight.root            |
| Per-variant outputs            | .../efficiencytool/results/Photon_final_bdt_ntbdtpair_{suffix}_{period}.root   |
| Nominal (1.5 mrad)             | .../efficiencytool/results/Photon_final_bdt_nom.root                           |
| Nominal (0 mrad)               | .../efficiencytool/results/Photon_final_bdt_0rad.root                          |